

AWS Orientation

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Agenda- Session 1

- What is the Cloud Computing?
- Types of Cloud Computing Deployment Models
- Benefits of Cloud Computing
- Virtualization
- Cloud computing service models
- AWS Global Infrastructure
- Key services overview:
 - Compute
 - Storage
 - Databases
 - Networking
 - Security

Agenda Session-2- AWS for AI & Data Science

- AWS AI/ML Services Overview
 - Amazon SageMaker - Complete ML platform
 - Amazon Rekognition - Computer vision
 - Amazon Comprehend - Natural language processing
 - Amazon Textract - Document analysis
 - Amazon Polly - Text-to-speech
 - Amazon Lex - Conversational AI
- Data Analytics and Storage Solutions
 - Amazon S3 - Object storage for data lakes
 - Amazon Redshift - Data warehousing
 - AWS Glue - ETL services
 - Amazon Kinesis - Real-time data streaming
 - Amazon Athena - Serverless query service

Agenda Session 3: Career Development & Learning Path

AWS Learning Resources (10 minutes)

- AWS Training and Certification portal
 - <https://aws.amazon.com/training/>
- AWS Educate program for students
 - <https://www.awseducate.com/student/s/content>
- Free tier benefits and limitations
 - <https://aws.amazon.com/blogs/aws/aws-free-tier-update-new-customers-can-get-started-and-explore-aws-with-up-to-200-in-credits/>
- AWS Skill Builder platform(AWS Builder ID)
 - <https://skillbuilder.aws>

AWS Certifications for AI/DS students (10 minutes)

- AWS Certified Cloud Practitioner (Foundational)
- AWS Certified Solutions Architect Associate (Intermediate)
- AWS Certified Machine Learning - Specialty
- AWS Certified Data Analytics - Specialty

Career Development & Learning Path cont..

Career Opportunities in Cloud & AI (10 minutes)

- **Job roles:** Cloud Architect, ML Engineer, Data Engineer
- Industry demand and salary expectations
- Building a portfolio with AWS projects
- AWS Skill Badges and digital credentials

Session 4: Hands-on

AWS Account Setup (10 minutes)

- Creating AWS Free Tier account
- Setting up Multi-Factor Authentication (MFA)
- Understanding billing and cost monitoring
- AWS Identity and Access Management (IAM) basics

AWS Console Navigation (10 minutes)

- Understanding regions and availability zones
- Introduction to AWS CLI (Command Line Interface)

Mini-Project 1: Launch EC2 Instance (15 minutes)

- Launching a Linux virtual server (Ubuntu/Amazon Linux)
- Selecting appropriate instance types
- Security groups and key pairs
- Connecting to the instance via SSH
- Basic server management commands

Mini-Project 2: Static Website Hosting (15 minutes)

What is the Cloud Computing?

- Cloud computing is the on-demand availability of computing resources (such as storage and infrastructure), as services over the internet.
- It eliminates the need for individuals and businesses to self-manage physical resources themselves, and only pay for what they use.

Types of cloud computing deployment models

- **Public clouds** are run by third-party cloud service providers.
- They offer compute, storage, and network resources over the internet, enabling companies to access shared on-demand resources based on their unique requirements and business goals.
- **Private clouds** are built, managed, and owned by a single organization and privately hosted in their own data centers, commonly known as “on-premises” or “on-prem.”
- **Hybrid clouds** combine public and private cloud models, allowing companies to leverage public cloud services and maintain the security and compliance capabilities commonly found in private cloud architectures.

Benefits of Cloud Computing

- Agility
- Elasticity
- Cost savings
- Deploy globally in minutes

Virtualization

- Virtualization is a technology that enables the creation of virtual environments from a single physical machine, allowing for more efficient use of resources by distributing them across computing environments.
- Using software, virtualization creates an abstraction layer over computer hardware, dividing a single system's components such as processors, memory, networks and storage into multiple virtual machines (VMs).
- Each VM runs its own operating system (OS) and behaves like a separate physical computer, despite sharing the same underlying hardware.

cloud computing service models

- The main cloud computing service models include **infrastructure as a service** offers compute and storage services,
- **platform as a service** offers a develop-and-deploy environment to build cloud apps, and
- **software as a service** delivers apps as services

AWS Global Infrastructure

Key services overview: Compute

- **Amazon Elastic Compute Cloud**
 - Complete control of your computing resources
 - Resizable compute capacity
 - Over 800+ types of compute instances
 - Instance type details
 - -General Purpose
 - Compute Optimized
 - -Memory optimized
 - -Accelerated optimized
 - -storage optimized
- Purchase options
- -On-demand instances
 - -saving plans
 - -spot instances

Key services overview: Storage

- Amazon S3
- Amazon EFS
- Amazon EBS

Key services overview: Databases

- Relational –Amazon RDS, Amazon Aurora,
- Non-relational-Amazon DynamoDB
- Non-relational document-Amazon DocumentDB
- Non-relational in-memory-Amazon ElastiCache
- Non-relational graph-Amazon Neptune
- Non-relational ledger-Amazon QLDB

Key services overview: Networking

- Amazon VPC(Virtual Private Cloud)
- Your private network space in the public AWS cloud
- Logical isolation for workloads
- Custom access controls and security settings

Key services overview: Security

- AWS IAM
- AWS IAM identity center
- Amazon Cognito
- AWS Organizations
- Amazon GuardDuty
- Many more
- <https://aws.amazon.com/products/security/?nc=sn&loc=2>